Chapter 3: Photosynthesis

Photosynthesis is the process by which green plants and some bacteria convert light energy into chemical energy. The process takes place mainly in the chloroplasts, which contain the green pigment chlorophyll. Chlorophyll absorbs light most strongly in the blue and red parts of the visible spectrum and reflects green light.

The overall reaction uses six molecules of carbon dioxide and six molecules of water to produce one molecule of glucose and six molecules of oxygen. Light energy drives the reaction, so the process is effectively powered by sunlight.

Photosynthesis has two main stages. The light-dependent reactions occur in the thylakoid membranes and split water to release oxygen. The Calvin cycle occurs in the stroma and uses ATP and NADPH from the light reactions to fix carbon dioxide into sugar.

Chapter 4: Cellular Respiration

Cellular respiration releases the chemical energy stored in glucose and makes it available to the cell as ATP. Most of this process takes place in the mitochondria, which are often described as the powerhouse of the cell.

Glycolysis splits one molecule of glucose into two molecules of pyruvate and produces a net gain of 2 ATP molecules. The Krebs cycle and the electron transport chain then extract much more energy, producing up to 36 ATP per glucose molecule under typical conditions.

Chapter 5: The Water Cycle

The water cycle describes how water moves continuously between the oceans, the atmosphere, and the land. Evaporation from oceans and lakes transfers water vapor into the air, where it cools and condenses to form clouds.

Precipitation returns water to the surface as rain, snow, or hail. Runoff carries water into rivers, and groundwater infiltration stores it underground for long periods.